

Mid Semester Examination

Algorithms Analysis and Design
IIIT Hyderabad, Monsoon 2025

September 25, 2025

There are 10 questions 10 marks each.

Maximum Marks: 100. Time: 90 min

1 × 10 = 10 marks

1. Fill in the following blanks.

1. The FFT of $(1, 0, 1, -1)$ is _____.
2. The worst-case complexity of deterministic quick-sort is _____.
3. The g.c.d. of two consecutive Fibonacci numbers is _____.
4. Solving the recurrence $T(n) = O(n) + \sum_{i=2}^{\lfloor \sqrt{n} \rfloor} T(\frac{n}{i^2})$, $T(\leq 1) = 1$, gives _____.
5. A Huffman code for aababcbcabcdabcdeabcdef is _____.
6. It is possible to find the median of n numbers in $O(\text{_____})$ time.
7. The edit-distance between SUNNY and SNOWY is _____.
8. An optimal parenthesization of a matrix-chain product with sequence of dimensions $\{5, 10, 3, 12, 5, 50, 6\}$ is _____.
9. The cut-property states that _____.
10. A language $A \subset \{0, 1\}^*$ is said to be mapping-reduced to language $B \subset \{0, 1\}^*$ if there exists a computable function f such that for all inputs in $\{0, 1\}^*$, _____.

2. Give an efficient algorithm to find the longest palindrome that is a subsequence of a given input string. Analyze and prove your answer. 10 marks

3. Design and analyze with proof an algorithm to compute the length of the longest increasing back-and-forth subsequence (alternatively increasing and decreasing) in an array. 10 marks

4. Given positive integers n and k , along with $p_1, \dots, p_n \in [0, 1]$, you wish to determine the probability of obtaining exactly k heads when n biased coins are tossed independently at random, where p_i is the probability that the i^{th} coin comes up heads. Give an efficient algorithm for this task. Suppose you wish to simulate an unbiased coin using all/some of the above coin(s), how would you go about it? 7 + 3 = 10 marks

5. Design an algorithm to make change (if possible) for n rupees using at most k coins of denominations in the set $\{x_1, x_2, \dots, x_n\}$ assuming infinite supply of each coin. If $x_i = c^i$, for some $c > 1$, show a greedy algorithm that yields an optimal solution. 6 + 4 = 10 marks

6. Design an algorithm to count the number of ways to construct sum n by throwing a dice one or more times. Each throw produces an outcome between 1 and 6. Modify the problem by adding a *cost* (associated with every solution) that you wish to optimize on such that (a) greedy strategy fails and (b) greedy works. Prove your answers, 4 + 3 + 3 = 10 marks

7. Define *matroids* and use a graphic matroid to solve the problem of minimum/maximum spanning tree using a greedy strategy. Give full details. 10 marks

8. You are given two multisets of integers, (a) A set of apples $A = \{a_1, a_2, \dots, a_n\}$ and (b) A set of bananas $B = \{b_1, b_2, \dots, b_m\}$, where each element lies in the range $[1, k]$. Define the pair sum frequency function $f : \{2, 3, \dots, 2k\} \rightarrow \mathbb{N}$ as $f(w) = |\{(a, b) \in A \times B \mid a + b = w\}|$. That is, $f(w)$ counts the number of pairs consisting of one apple and one banana whose weights sum to exactly w . Design and analyze an efficient algorithm (using FFT that computes discrete convolutions efficiently) to compute $f(w)$ for all $w \in \{2, \dots, 2k\}$. 10 marks

9. Prove the following: $2^{\frac{1}{2}} \times 4 = 10$ marks

1. the set of all subsets of $\mathbb{N}$ is uncountable.

2. A_{TM} is undecidable.

3. $\overline{A_{TM}}$ is unrecognizable.

4. A_{DFA} is decidable.

10. Design and analyze (both efficacy/correctness/optimality and efficiency/complexity) an algorithm that computes the longest common subsequence (LCS) of two sequences. Execute your algorithm on some example of your choice. 5 + 2 + 1 + 2 = 10 marks.

ALL THE BEST!